Naïve Bayes, Nearest Neighbours and Cost Functions

3

Pre

CS 3244
Machine Learning

NUS National University of Singapore | Computing

Recap and Forecast

Concept learning as search through $\mathcal{H}$

General-to-specific ordering over $\mathcal{H}$

Candidate elimination algorithm

Boundaries S and G characterize learner's uncertainty

Active learner can generate informative queries

Stronger inductive bias allows classification of greater proportion of unobserved input instances

Topics for this week

- Bayes' rule and Naïve Bayes
- Distance metrics and k Nearest Neighbours
- Training and testing times for both

- Smoothing
- Cost (Loss) Functions
- Noisy Targets
- Curse of Dimensionality

Naive Bayes

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Bayes' Rule

$$P(y|X) = \frac{P(X|y)P(y)}{P(X)}$$

Bayes' Rule

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Bayes' Rule

Likelihood: How probable is the data given that our class is positive?

Prior: How probable is an instance to be positive without seeing any data?

$$P(y|X) = \frac{P(X|y)P(y)}{P(X)}$$

Posterior: How probable the instance a positive case?

Marginal: How probable is the evidence under both classes of instances (both positive or negative)

Probability Chain/Product Rule

$$P(x) = P(x_1, \dots, x_n)$$

We can always write

$$P(x_1, \dots, x_n) = P(x_1 | x_2, \dots x_n)\, P(x_2, \dots, x_n)$$

Repeatedly applying this idea, we can write

$$= P(x_1 | x_2, \dots, x_n) P(x_2 | x_3, \dots, x_n) P(x_3 | x_4, \dots x_n) \dots P(x_n)$$

This factorization holds for any ordering of the features.

Naïve Bayes

Likelihood: How probable are the data features given that our class is positive and assuming they are **conditionally independent**?

Prior: How probable is an instance to be positive without seeing any data?

$$P(y|x) = P(x_1|y) \times P(x_2|y) \times \cdots \times P(x_n|y) \times P(y)$$

Posterior: How probable the instance a positive case?

Marginal: To think about. Where did it go?

Trace of Naïve Bayes

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Weather Dataset

Outlook	Temperature	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Frequency Table

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3

Outlook	Temperature	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Likelihood Table

		Play Golf		
		Yes	No	
Outlook	Sunny	3/9	2/5	5/14
	Overcast	4/9	0/5	4/14
	Rainy	2/9	3/5	5/14
		9/14	5/14	

Likelihood Table

$$P(x_i|y) = P(Sunny|Yes) = \frac{3}{9}$$

$$P(x_i) = P(Sunny) = \frac{5}{14}$$

		Play Golf		
		Yes	No	
Outlook	Sunny	3/9	2/5	5/14
	Overcast	4/9	0/5	4/14
	Rainy	2/9	3/5	5/14
		9/14	5/14	

$$P(y) = P(Yes) = \frac{9}{14}$$

$$P(y|x) = P(Yes|Sunny) = \frac{3}{9} \times \frac{9}{14} \div \frac{5}{14} = \frac{3}{5}$$

All of the likelihood tables

These statistics make up the model representation for NB, our h_θ.

		Play Golf	
		Yes	No
Outlook	Sunny	3/9	2/5
	Overcast	4/9	0/5
	Rainy	2/9	3/5

		Play Golf	
		Yes	No
Temperature	Hot	2/9	2/5
	Mild	4/9	2/5
	Cool	3/9	1/5

		Play Golf	
		Yes	No
Windy	True	3/9	3/5
	False	6/9	2/5

		Play Golf	
		Yes	No
Humidity	High	3/9	4/5
	Normal	6/9	1/5

Applying NB at test time

Outlook	Temperature	Humidity	Windy	Play Golf
Rainy	Cool	High	True	?

Applying NB at test time

Outlook	Temperature	Humidity	Windy	Play Golf
Rainy	Cool	High	True	?

$$P(Yes|x) = P(Rainy|Yes) \times P(Cool|Yes) \times P(High|Yes) \times P(True|Yes) \times P(Yes)$$

$$P(No|x) = P(Rainy|No) \times P(Cool|No) \times P(High|No) \times P(True|No) \times P(No)$$

		Play Golf	
		Yes	No
Outlook	Sunny	3/9	2/5
	Overcast	4/9	0/5
	Rainy	2/9	3/5

		Play Golf	
		Yes	No
Temperature	Hot	2/9	2/5
	Mild	4/9	2/5
	Cool	3/9	1/5

		Play Golf	
		Yes	No
Humidity	High	3/9	4/5
	Normal	6/9	1/5

		Play Golf	
		Yes	No
Windy	True	3/9	3/5
	False	6/9	2/5

Applying NB at test time

Outlook	Temperature	Humidity	Windy	Play Golf
Rainy	Cool	High	True	?

$$P(Yes|x) = P(Rainy|Yes) \times P(Cool|Yes) \times P(High|Yes) \times P(True|Yes) \times P(Yes)$$

$$P(Yes|x) = \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} = 0.00529$$

$$P(No|x) = P(Rainy|No) \times P(Cool|No) \times P(High|No) \times P(True|No) \times P(No)$$

$$P(No|x) = \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} = 0.02057$$

		Play Golf	
		Yes	No
Outlook	Sunny	3/9	2/5
	Overcast	4/9	0/5
	Rainy	2/9	3/5

		Play Golf	
		Yes	No
Temperature	Hot	2/9	2/5
	Mild	4/9	2/5
	Cool	3/9	1/5

$$P(Yes|x) = P(Rainy|Yes) \times P(Cool|Yes) \times P(High|Yes) \times P(True|Yes) \times P(Yes)$$

$$P(Yes|x) = \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} = 0.00529$$

Applying NB at test time

Outlook	Temperature	Humidity	Windy	Play Golf
Rainy	Cool	High	True	?

$$P(Yes|x) = \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} = 0.00529$$

$$P(No|x) = \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} = 0.02057$$

		Play Golf	
		Yes	No
Outlook	Sunny	3/9	2/5
	Overcast	4/9	0/5
	Rainy	2/9	3/5

		Play Golf	
		Yes	No
Temperature	Hot	2/9	2/5
	Mild	4/9	2/5
	Cool	3/9	1/5

		Play Golf	
		Yes	No
Humidity	High	3/9	4/5
	Normal	6/9	1/5

		Play Golf	
		Yes	No
Windy	True	3/9	3/5
	False	6/9	2/5

Applying NB at test time

Outlook	Temperature	Humidity	Windy	Play Golf
Rainy	Cool	High	True	?

$$P(Yes|x) = \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} = 0.00529 \qquad \frac{0.00529}{0.02057 + 0.00529} = 0.2$$

$$P(No|x) = \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} = 0.02057 \qquad \frac{0.02057}{0.02057 + 0.00529} = 0.8$$

		Play Golf	
		Yes	No
Outlook	Sunny	3/9	2/5
	Overcast	4/9	0/5
	Rainy	2/9	3/5

		Play Golf	
		Yes	No
Temperature	Hot	2/9	2/5
	Mild	4/9	2/5
	Cool	3/9	1/5

		Play Golf	
		Yes	No
Humidity	High	3/9	4/5
	Normal	6/9	1/5

		Play Golf	
		Yes	No
Windy	True	3/9	3/5
	False	6/9	2/5

Applying NB at test time

Outlook	Temperature	Humidity	Windy	Play Golf
Rainy	Cool	High	True	No

$$P(Yes|x) = \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} = 0.00529 \qquad \frac{0.00529}{0.02057 + 0.00529} = 0.2$$

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max

		Play Golf	
		Yes	No
Outlook	Sunny	3/9	2/5
	Overcast	4/9	0/5
	Rainy	2/9	3/5

		Play Golf	
		Yes	No
Temperature	Hot	2/9	2/5
	Mild	4/9	2/5
	Cool	3/9	1/5

		Play Golf	
		Yes	No
Humidity	High	3/9	4/5
	Normal	6/9	1/5

		Play Golf	
		Yes	No
Windy	True	3/9	3/5
	False	6/9	2/5

Summary – Naïve Bayes

Bayesian approaches deal with uncertainty in a principled way: Probabilistically

Naïve Bayes makes a very strong assumption that features are conditionally independent of each other

Has some fixable problems

- Doesn't handle zeros
- Only works for categorical predictions, not regression

Nearest Neighbour

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Analogizer

Compare something with something else to assist understanding

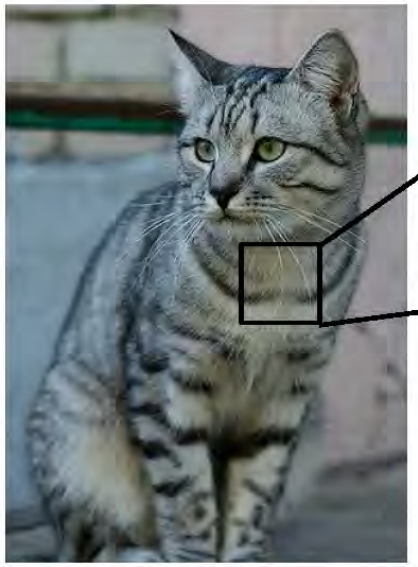
To classify a test instance, assign it to its closest neighbor

Train: Memorize the training data

Test: Compute which training instance is closest to the test instance

Recap: image representation

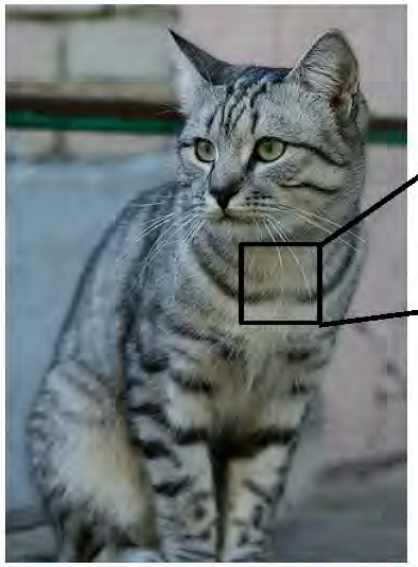
The semantic gap: We see a cat…



```
[[105 112 108 111 104  99 106  99  96 103 112 119 104  97  93  87]
 [ 91  98 102 106 104  79  98 103  99 105 123 136 110 105  94  85]
 [ 76  85  90 105 128 105  87  96  95  99 115 112 106 103  99  85]
 [ 99  81  81  93 120 131 127 100  95  98 102  99  96  93 101  94]
 [106  91  61  64  69  91  88  85 101 107 109  98  75  84  96  95]
 [114 108  85  55  55  69  64  54  64  87 112 129  98  74  84  91]
 [133 137 147 103  65  81  80  65  52  54  74  84 102  93  85  82]
 [128 137 144 140 109  95  86  70  62  65  63  63  60  73  86 101]
 [125 133 148 137 119 121 117  94  65  79  80  65  54  64  72  98]
 [127 125 131 147 133 127 126 131 111  96  89  75  61  64  72  84]
 [115 114 109 123 150 148 131 118 113 109 100  92  74  65  72  78]
 [ 89  93  90  97 108 147 131 118 113 114 113 109 106  95  77  80]
 [ 63  77  86  81  77  79 102 123 117 115 117 125 125 130 115  87]
 [ 62  65  82  89  78  71  80 101 124 126 119 101 107 114 131 119]
 [ 63  65  75  88  89  71  62  81 120 138 135 105  81  98 110 118]
 [ 87  65  71  87 106  95  69  45  76 130 126 107  92  94 105 112]
 [118  97  82  86 117 123 116  66  41  51  95  93  89  95 102 107]
 [164 146 112  80  82 120 124 104  76  48  45  66  88 101 102 109]
 [157 170 157 120  93  86 114 132 112  97  69  55  70  82  99  94]
 [130 128 134 161 139 100 109 118 121 134 114  87  65  53  69  86]
 [128 112  96 117 150 144 120 115 104 107 102  93  87  81  72  79]
 [123 107  96  86  83 112 153 149 122 109 104  75  80 107 112  99]
 [122 121 102  80  82  86  94 117 145 148 153 102  58  78  92 107]
 [122 164 148 103  71  56  78  83  93 103 119 139 102  61  69  84]]
```

Recap: image representation

The semantic gap: We see a cat…



```
[[105 112 108 111 104  99 106  99  96 103 112 119 104  97  93  87]
 [ 91  98 102 106 104  79  98 103  99 105 123 136 110 105  94  85]
 [ 76  85  90 105 128 105  87  96  95  99 115 112 106 103  99  85]
 [ 99  81  81  93 120 131 127 100  95  98 102  99  96  93 101  94]
 [106  91  61  64  69  91  88  85 101 107 109  98  75  84  96  95]
 [114 108  85  55  55  69  64  54  64  87 112 129  98  74  84  91]
 [133 137 147 103  65  81  80  65  52  54  74  84 102  93  85  82]
 [128 137 144 140 109  95  86  70  62  65  63  63  60  73  86 101]
 [125 133 148 137 119 121 117  94  65  79  80  65  54  64  72  98]
 [127 125 131 147 133 127 126 131 111  96  89  75  61  64  72  84]
 [115 114 109 123 150 148 131 118 113 109 100  92  74  65  72  78]
 [ 89  93  90  97 108 147 131 118 113 114 113 109 106  95  77  80]
 [ 63  77  86  81  77  79 102 123 117 115 117 125 125 130 115  87]
 [ 62  65  82  89  78  71  80 101 124 126 119 101 107 114 131 119]
 [ 63  65  75  88  89  71  62  81 120 138 135 105  81  98 110 118]
```

But the computer just sees a grid of numbers.

Here we have a color image with 3 channels (R,G,B) and 256 levels of intensity.

Width × Height × Channels
800 × 600 × 3 = 1.44M integers

CIFAR 10 Dataset

10 classes

10K test instances

50K training instances

Distance Metric

The key to determining the meaning of "nearest"

L1 distance metric: $d\left(x^{(*)}, x^{(1)}\right) = \sum_{p} \left| x_p^{(*)} - x_p^{(1)} \right|$

Distance Metric

The key to determining the meaning of "nearest"

L1 distance metric: $d\left(x^{(*)}, x^{(1)}\right) = \sum_{p} \left| x_p^{(*)} - x_p^{(1)} \right|$

test image

56	32	10	18
90	23	128	133
24	26	178	200
2	0	255	220

-

training image

10	20	24	17
8	10	89	100
12	16	178	170
4	32	233	112

=

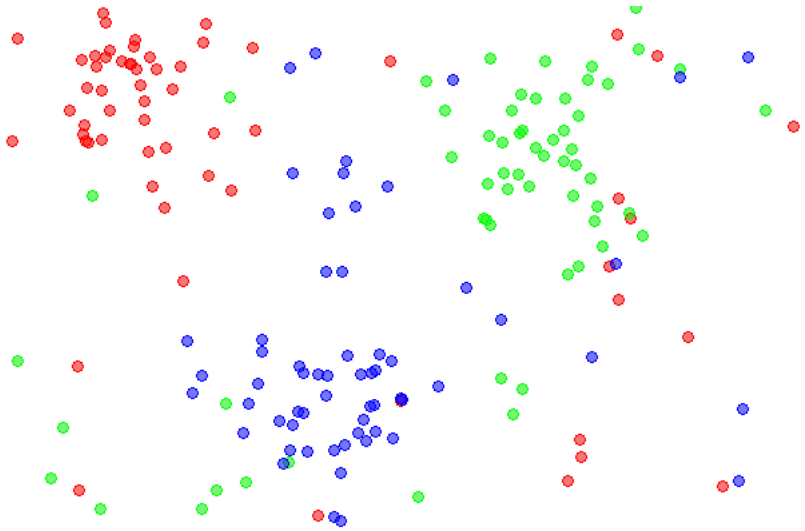
pixel-wise absolute value differences

46	12	14	1
82	13	39	33
12	10	0	30
2	32	22	108

$\xrightarrow{\text{add}}$ 456

So what does it look like?

Here's a three class problem, plotted using just two input features:



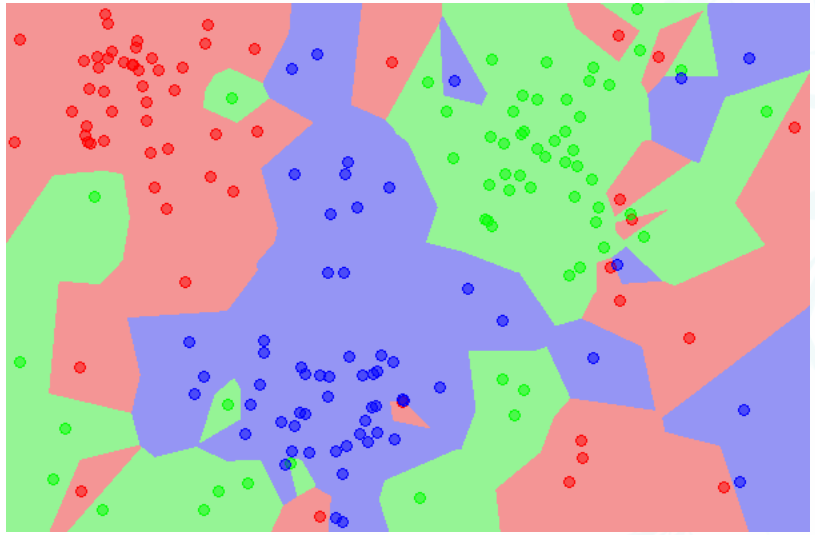
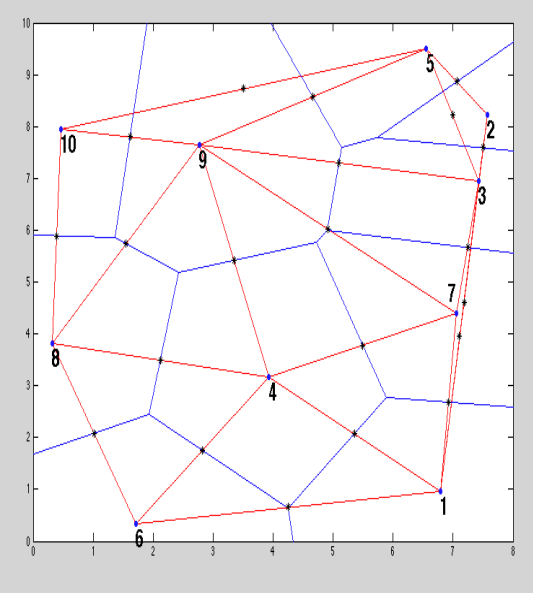
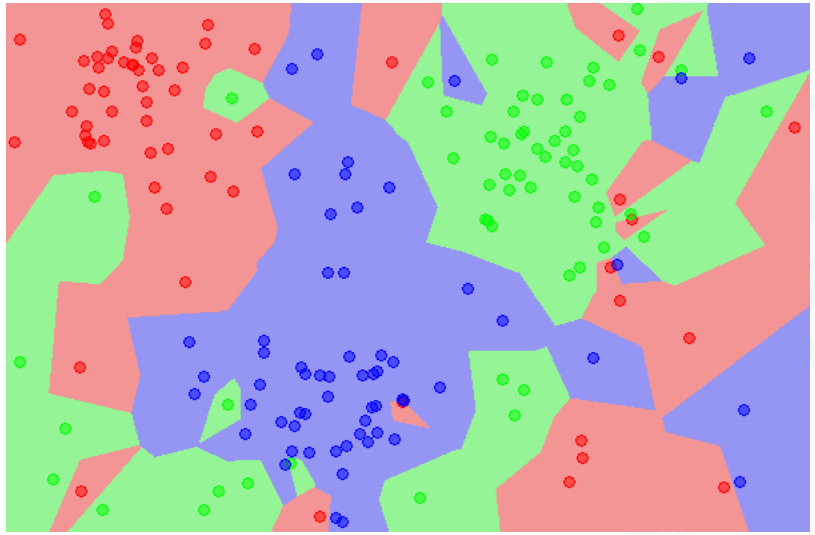


Photo credits: Agor153 - Own work, CC BY-SA 3.0

So what does it look like?

Here's a three class problem, plotted using just two input features:





The decision boundaries form a Voronoi tessellation.

Photo credits: Agor153 - Own work, CC BY-SA 3.0

Sample output kNN on CIFAR-10

Summary – Nearest Neighbor

A parameter free-algorithm (non-parametric)

Requires an immense amount of time to apply (needs to cycle through all of the test data)

How can we speed this up?

Doesn't work well in high dimensions (why?)

Cost Functions

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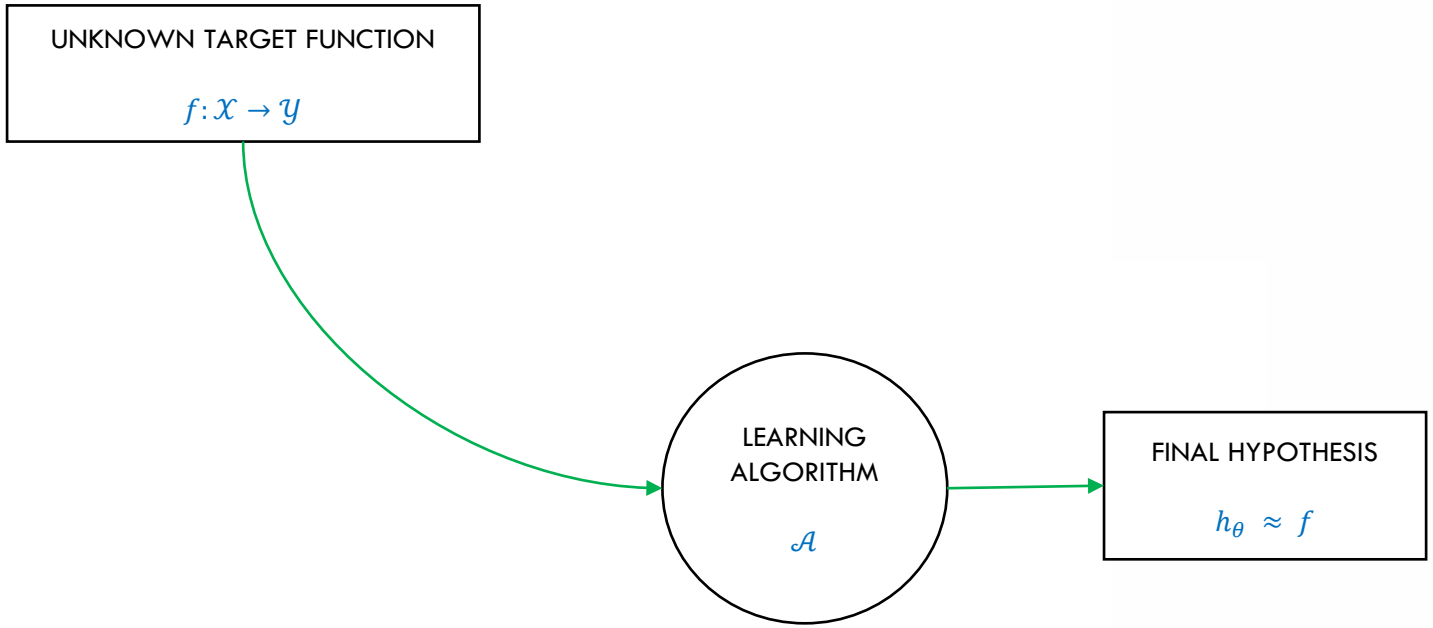
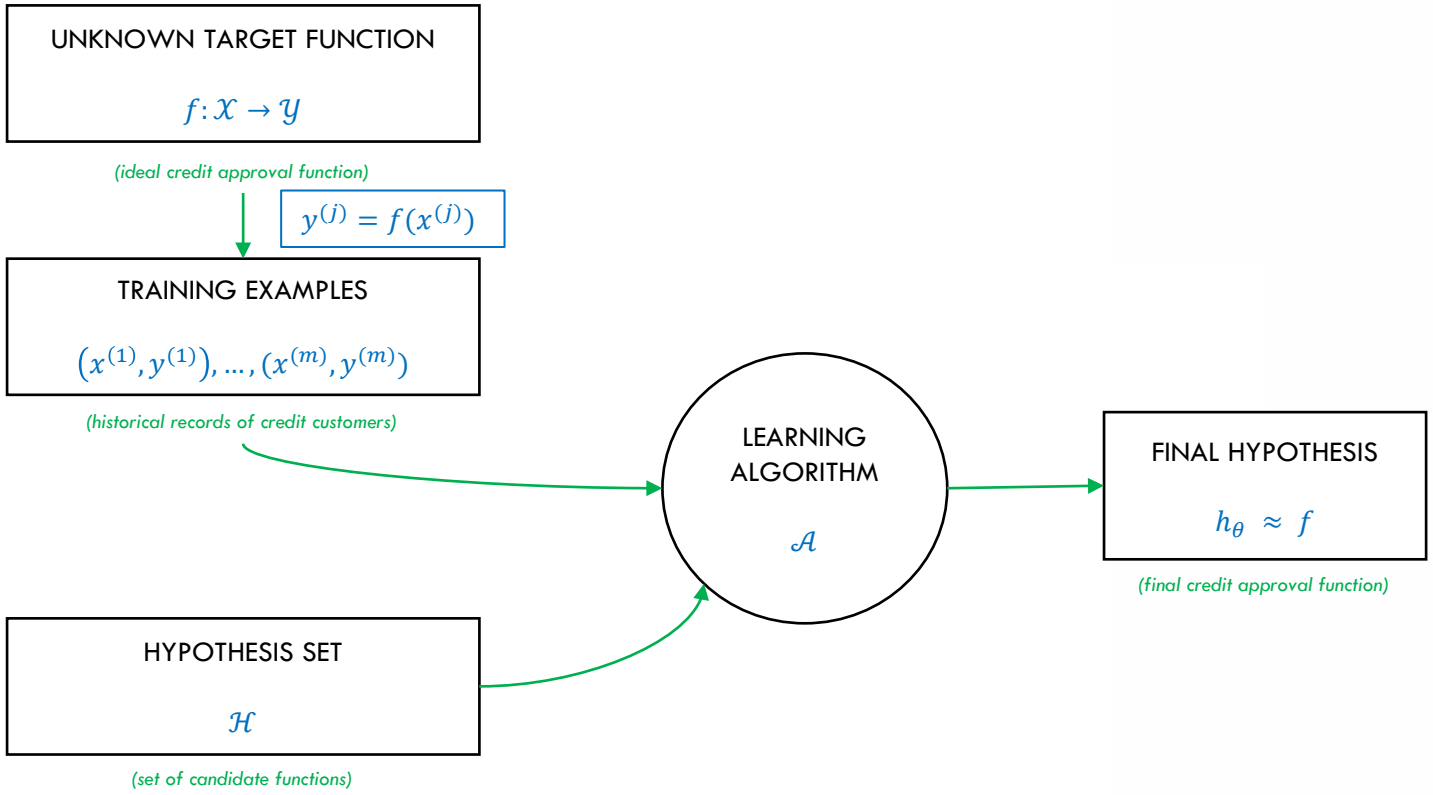
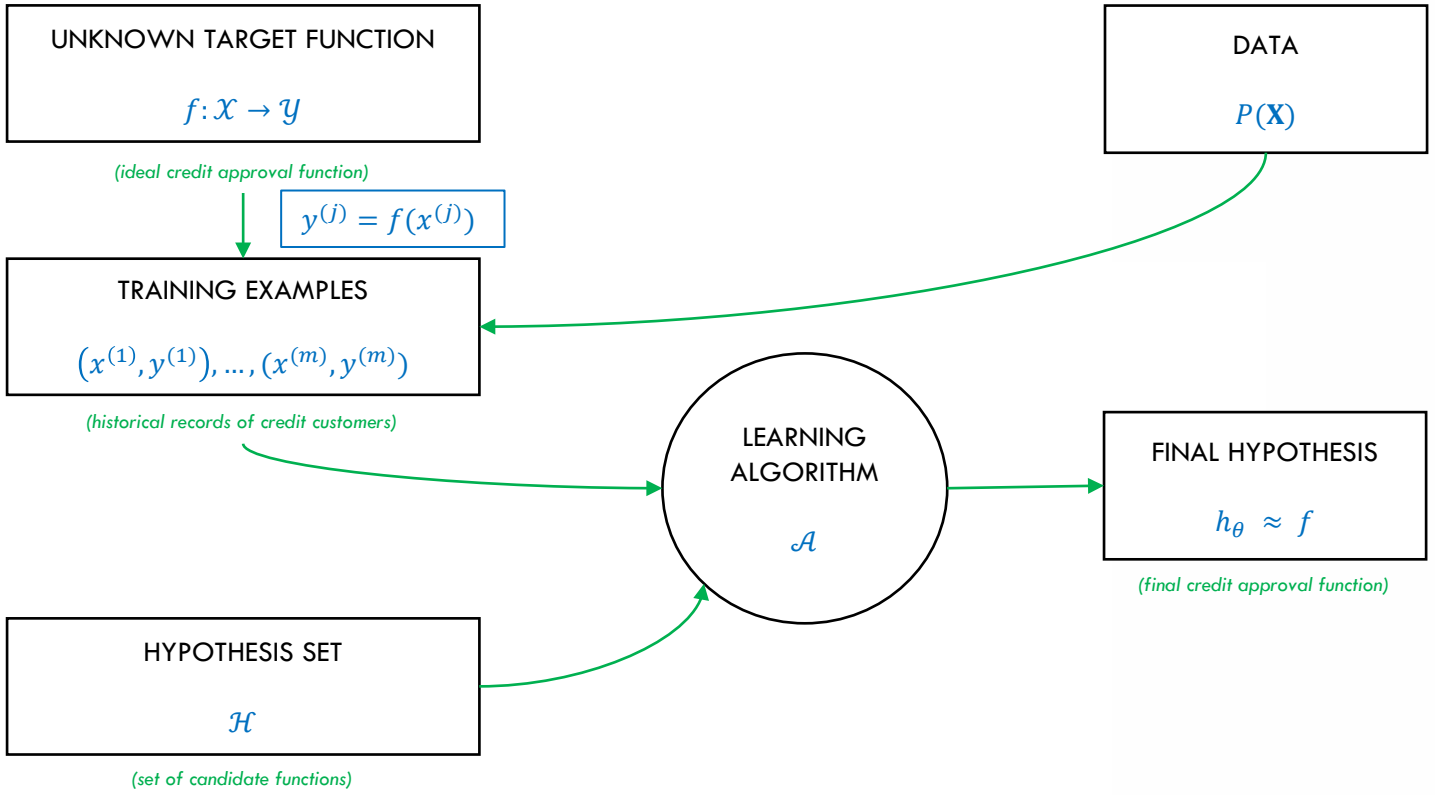
The learning diagram



The learning diagram



The learning diagram, with data



Cost functions

What does $h_\theta \approx f$ mean?

Need a cost function $J(h_\theta, f)$

L = loss

This is almost always a pointwise definition: $j(h_\theta(x), f(x))$

Examples we've seen:

Squared error $(h_\theta(x) - f(x))^2$

Binary error $[[h_\theta(x) \neq f(x)]]$

Which is for classification?

if true → 1
else → 0

From pointwise to overall

Overall cost $J(h, f) =$

average of pointwise cost $j(h(x), f(x))$

Training cost:

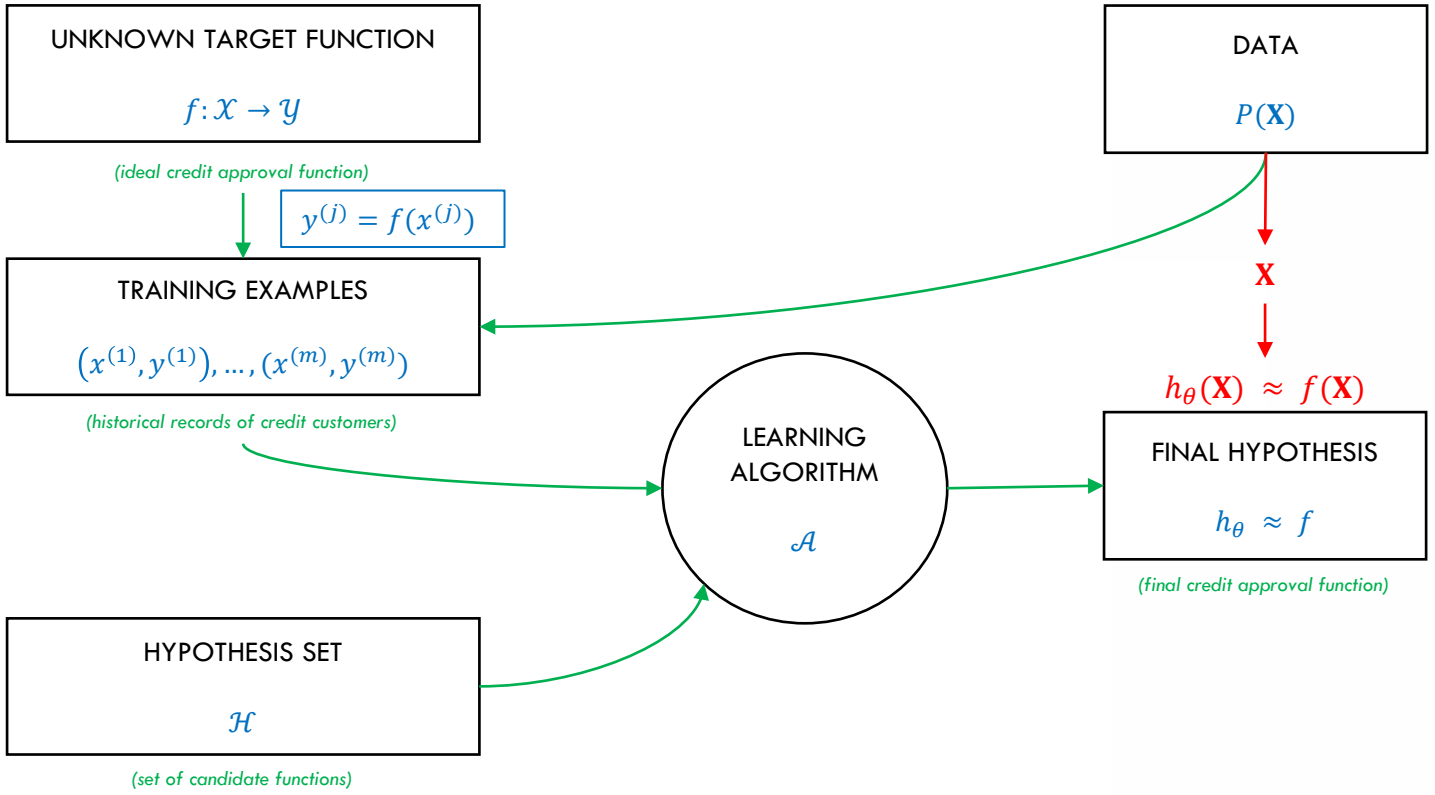
$$J_{train} = \frac{1}{m}\sum_{j=1}^{m} j(h_\theta(x^{(j)}), f(x^{(j)}))$$

Why not a sum instead of an average?

Test cost:

$$J_{test} = \mathbb{E}_x[j(h_\theta(x), f(x))]$$

The learning diagram with testing data, pointwise error



Choosing your cost function

Are you sick?

Two types of costs:

false positive (accept) or

false negative (reject)

How should we penalize for each type?

$$f \longrightarrow \begin{cases} +1 & \text{sick} \\ -1 & \text{well} \end{cases}$$

h_θ	f: $+1$	f: -1
$+1$	True Positive	False Positive
-1	False Negative	True Negative

During your last final exam before graduation

Are you sick?

False negative – get better on your own, or come back to the clinic later. At least you graduate on time.

False positive – Take the exam next year. Possibly pay tuition fees. $$$

$$f \longrightarrow \begin{cases} +1 & \text{sick} \\ -1 & \text{well} \end{cases}$$

h_θ \ f	$+1$	-1
$+1$	0 True Positive	100 False Positive
-1	1 False Negative	0 True Negative

During COVID-19 Pandemic

Are you sick?

False negative highly costly!
People and the economy dies.

False positive requires
inconvenience of quarantine.

$$f \longrightarrow \begin{cases} +1 & \text{sick} \\ -1 & \text{well} \end{cases}$$

h_θ \ f	$+1$	-1
$+1$	0 True Positive	1 False Positive
-1	10,000 False Negative	0 True Negative

How you measure matters

Where possible, we should use cost functions that fit the task, specified by the user.

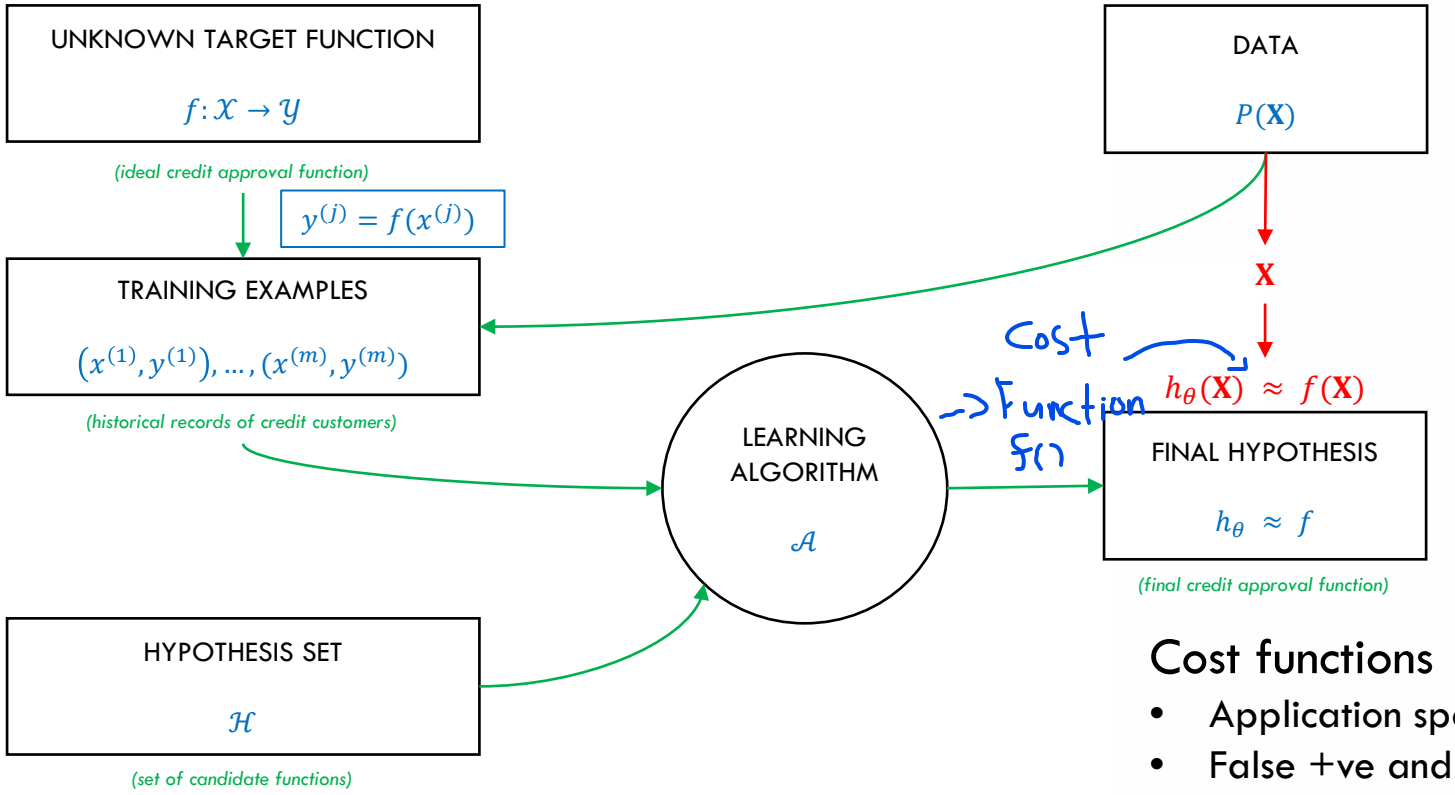
However, this isn't always possible. Then use:

Plausible measures: squared error ≡ Gaussian noise

Convenient measures: closed form solution, convex optimization

The sweet spot: Plausible and Convenient

Summary – the learning diagram with testing data, pointwise error



Quick Question: where does the cost function go?

Cost functions

- Application specific, user should specify
- False +ve and –ve may differ in cost